Abdullah Korra

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Summary

Hands-on electrical & robotics engineering with end-to-end build experience: Arduino to Raspberry Pi serial communication with Python GUIs. Build an autonomous farming robot system. Strong in CAD fabrication designs, Strong in testing, documentation, and turning prototypes into repeatable systems. Seeking an entry-level role in CAD design/Electrical Engineering (EE)/Robotics/Semiconductors/Energy/Data Science, Manufacturing settings.

Education

The College of Idaho — **Caldwell, ID**
B.S. in Math-Physics — *Expected May 2026*
Minors: Computer Science, History, Art
Honors: Full-Tuition Scholarship, Gipson Honors Program
Relevant Topics: embedded systems, controls, power electronics, signals, data structures

UWC Adriatic — **Duino, Italy**
International Baccalaureate Diploma — *2019–2021*

Core Skills

Fabrication: CAD (Fusion 360), 3D printer; Powermatic table saw/bandsaw/drill press; belt/disc sander.

Electrical: Soldering, building and Testing breadboard Circuits; Arduino and Raspberry Pi base projects wiring.

Software: Python (Tkinter/ttkbootstrap, Kivy, NumPy, pandas, matplotlib, Astropy, PyVista); Arduino; Raspberry Pi: R; LaTeX; ROS2 (basic); Linux (systemd user services).

Tools: GitHub, Jupyter Lab, VS Code, oscilloscope, Electronic Test Equipment, Bash, communication (reading/writing technical docs and presentations), Microsoft Office Suite.

Experience

Personal Project: AgriBot Autonomous Farming Robot — *2024–2025*
- Engineered motor subsystem for 4 independent drive wheels + plow using BTS7960 H-bridges; wrote Arduino firmware with open-loop PWM, A-channel interrupt encoders, and ASCII command protocol from the Pi.
- Pi GUI (Python + ttkbootstrap) in kiosk mode via user systemd service & autostart wrapper.
- Built boundary & path planning: lat/lon→XY projection and implement dashed route visualization.
- Integrated GPS (NEO-6M), IMU (MPU-6050), LaRA module, and ultrasonic-servo for device-to-device communication.

The College of Idaho — **Caldwell, ID**
Undergraduate Astronomy Research Assistant — *2023–Present*
- Processed and analyzed large astronomy datasets using astro-pyvista.
- Developed a Python pipeline producing science-ready 1D/2D spectra for multi-slit spectroscopy.
- Delivered a Jupyter/ipywidgets GUI guiding users through the spectroscopy data reduction.

Others — **The Gambia & Italy**
Car Mechanic Apprentice — *2012–2016, 2019*
Hands-on experience in automotive repair and maintenance.

Projects

- **Photoelectric Effect Experiment:** Designed and assembled an LED-based photoelectric effect apparatus.
- **Ionic Thruster:** Designed and built a functional ionic thruster.
- **LED Art Project:** Built an LED art circuit, comparing functionality with and without an IC.
- **KOSMOS Multi-Slit Spectroscopy Reduction:** Developed a step-by-step data-reduction tutorial.

Presentations & Awards

- **Poster Presenter (2023-2025):** M.J. Murdock College Science Research Conference.
- **Poster Presenter (2025):** Idaho Conference on Undergraduate Research (ICUR).
- **Award:** UWC-Shelby Davis Scholar.